



IILM UNIVERSITY

Department of Computer Science & Engineering

PRODUCT ENGINEERING INTERNSHIP

Internship at MarTechAdda

A Product Engineering internship spanning CRM finance workflows and the Ledgerly billing platform.



STUDENT

Anubhav Sachan



ROLL NUMBER

2410030896



PROGRAM

B.Tech (CSE), Cyber Security



BATCH

2024 – 28



ORGANIZATION

MarTechAdda Pvt. Ltd.



DURATION

15 Jun – 31 Jul 2026

INTERNSHIP OVERVIEW

Internship & Organization Overview



Organization Environment

MarTechAdda Pvt. Ltd. operates a product-development environment focused on business-facing web applications spanning CRM, finance, and billing operations. As a Product Engineering Intern, the work spanned the product lifecycle — from UI development to backend integration — within a collaborative, Git-based engineering workflow.



ROLE

Product Engineering Intern



DURATION

15 Jun 2026 – 31 Jul 2026



SUPERVISOR

Ritesh Kumar

FOCUS AREAS



Product development



CRM workflows



Finance workflows



Billing software



Backend integration



Application security

SCOPE OF WORK

Objectives & Technology Stack

INTERNSHIP OBJECTIVES



Gain practical product-engineering experience



Strengthen React.js, TypeScript, Node.js & Express.js skills



Develop and contribute to the CRM Finance module



Design a structured, auditable refund lifecycle



Build billing & invoicing features for Ledgerly



Identify and resolve application security vulnerabilities

TECHNOLOGY STACK



Frontend

React.js · TypeScript · Tailwind CSS · Vanilla CSS



Backend

Node.js · Express.js



Database

PostgreSQL · Prisma ORM · MongoDB



Auth & Security

Passport.js · Google OAuth 2.0



Tooling

Git & GitHub · LLM-assisted dev tools

CORE CONTRIBUTION

CRM Finance Module



OWNED BY ME

Transaction Dashboard

Primary ownership — ledger view of cash flow, expenditure & customer credit



Revenue Dashboard

Structured visibility into overall business revenue



Refund Tracking

End-to-end refund lifecycle management



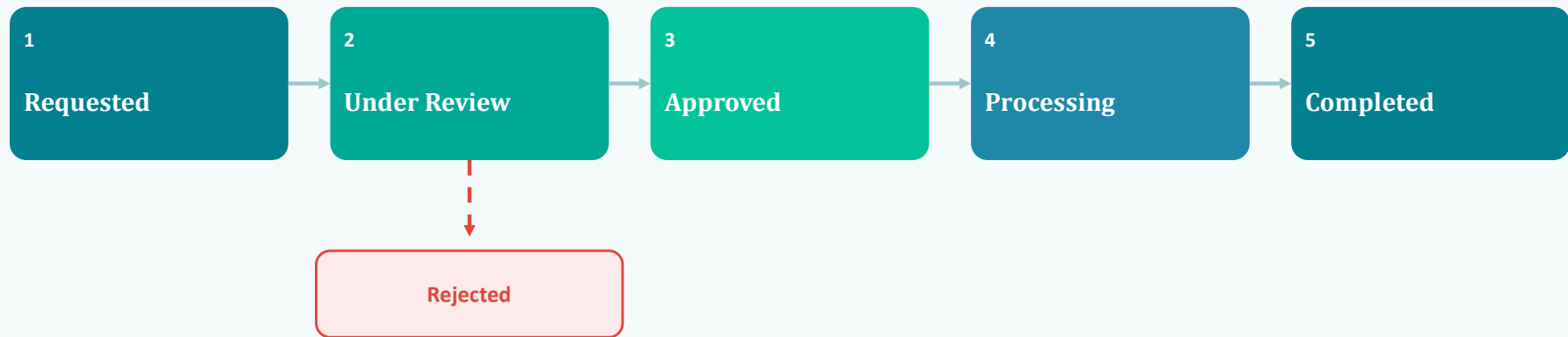
Payout Tracking

Tracking of outgoing payouts within the CRM

The Transaction Dashboard provided a structured ledger view for tracking cash flow, expenditure, and customer credit — giving the finance team clear, consolidated financial visibility across CRM operations.

REFUND TRACKING

Refund Lifecycle



The workflow provides structured and traceable refund status management — moving each request through clearly defined stages from initial request to final resolution.

ENGINEERING DEPTH

Backend Integration & Security

BACKEND INTEGRATION

-  Express.js server integration
-  Database model integration
-  Global state management
-  Synchronization issue resolution
-  Route protection
-  Authorization-related improvements
-  Development database isolation

SECURITY ISSUES IDENTIFIED & RESOLVED



Issue 1 — Unguarded Routes

Certain routes lacked protection, creating a risk of unauthorized data exposure.



Issue 2 — Editable Frontend Role Field

The user role field was editable on the frontend, posing a privilege-escalation risk.



Resolved by restricting access based on verified user roles rather than editable client-side data.

PRODUCT DEVELOPMENT

Ledgerly — Billing & Invoicing Platform



Invoices



Estimates / Quotations



Clients



Catalog Profiles



Client Ledgers



Transaction Records



Activity Timelines



Contacts



Billing workflow management



Client information management



Printable financial documents



CGST / SGST tax breakdown



Quotation-to-invoice conversion

LEDGERLY

Ledgerly Workflow

INVOICE LIFECYCLE



QUOTATION CONVERSION



SUPPORTING FEATURES



CGST / SGST Tax Breakdown

Automatic tax split shown on every invoice



Printable Financial Documents

Invoices and estimates generated as print-ready documents



Server-side Pagination

15 items displayed per page across listing views

TIMELINE

Week-wise Internship Journey



WEEK 1-2

Development Foundations

- Git / GitHub
- Repository structure
- Branches & pull requests
- React/Node/Express/DB basics



WEEK 3

UI Designing

- Blog page
- Resources page
- Responsive interfaces
- Search & category filters



WEEK 4

CRM Finance

- Transaction dashboard
- Revenue dashboard
- Refund tracking
- Payout tracking



WEEK 5

Integration & Security

- Backend / DB integration
- State management
- Security fixes
- Data integrity



WEEK 6

Ledgerly & Portfolio

- Billing software
- Invoices & estimates
- Client management
- Personal portfolio

CONCLUSION

Learning Outcomes & Conclusion

- ✓ Practical full-stack product engineering experience
- ✓ Billing and invoicing software development
- ✓ Authentication and authorization exposure
- ✓ Git / GitHub collaborative development
- ✓ Understanding of real-world business workflows
- ✓ CRM financial workflow development
- ✓ Backend and database integration
- ✓ Security vulnerability identification & resolution
- ✓ Debugging and integration experience

Thank You

Questions & Discussion

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